UNIT 4 — Important Exam Notes

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1. Arithmetic Circuit

Definition

Arithmetic Circuit ek digital circuit hota hai jo different **arithmetic operations** perform karta hai, jaise:

- Addition
- Subtraction
- Increment
- Decrement
- Transfer

Arithmetic circuit mainly **binary numbers** par operations perform karta hai.

A basic arithmetic circuit mein generally:

- **Full Adder**
- **Multiplexers**
- **Logic gates**

use hote hain.

Basic idea

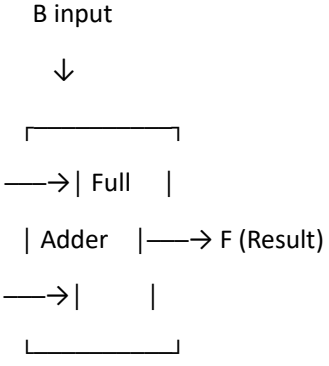
Arithmetic circuit mein ek binary number **A**, ek input **B**, aur **selection/control lines** hoti hain. Control input decide karta hai ki circuit ko kaunsa arithmetic operation perform karna hai.

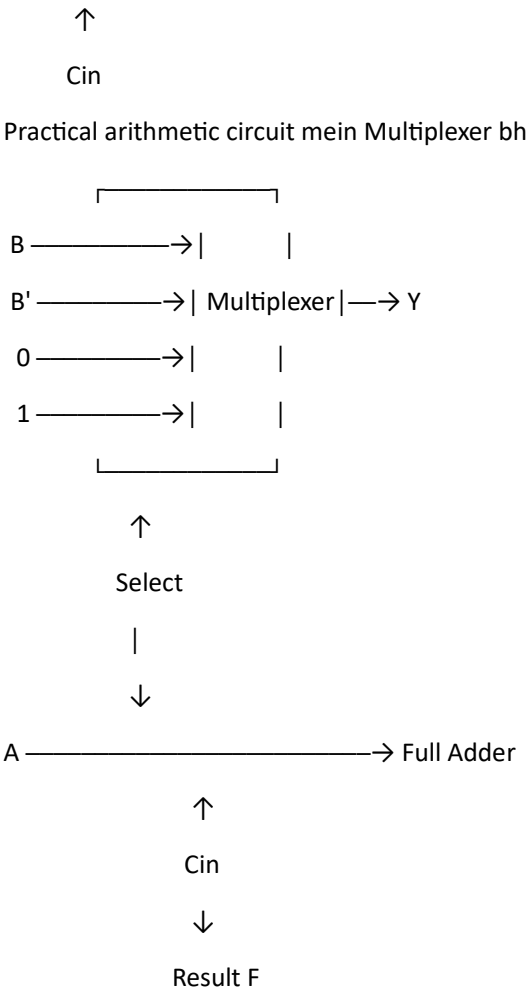
Arithmetic Circuit using Full Adder

Ek **n-bit arithmetic circuit** banane ke liye generally n full adders use kiye ja sakte hain.

For example, agar 4-bit numbers par operation karna hai, to **4 Full Adders** use honge.

Basic Diagram





Practical arithmetic circuit mein Multiplexer bhi use hota hai:

Multiplexer different inputs mein se ek input select karta hai, aur Full Adder us selected value ko **A aur Carry input** ke saath add karta hai.

Arithmetic Circuit Operations

A commonly used arithmetic circuit can perform operations such as:

Select	Operation
000	**F = A**
001	**F = A + 1**
010	**F = A + B**
011	**F = A + B + 1**
100	**F = A + B'**
101	**F = A + B' + 1**

Select Operation

110 **F = A - 1**

111 **F = A** / transfer-type operation

Exam mein apne teacher/textbook ki exact operation table follow karna, kyunki selection codes circuit design ke according vary kar sakte hain.

Important Arithmetic Operations

1. Addition

Two binary numbers ko add karta hai.

F = A + B

Example:

1010

+ 0011

1101

So,

10 + 3 = 13

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2. Increment

Number mein **1 add** karta hai.

F = A + 1

Example:

A = 0101

A + 1 = 0110

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3. Subtraction

Subtraction generally **2's complement method** se perform ki ja sakti hai.

Formula:

A - B = A + 2's complement of B

Example:

A = 0110

B = 0011

2's complement of B = 1101

0110

+ 1101

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1 0011

Final result:

0011 = 3

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4. Decrement

Number mein se **1 subtract** karta hai.

F = A - 1

Example:

A = 1000

A - 1 = 0111

Why Arithmetic Circuit is Important?

- CPU mein arithmetic operations perform karne ke liye use hota hai.
- ALU ka important part hota hai.
- Addition/subtraction jaise operations efficiently perform karta hai.
- Digital computers ke internal calculations mein important role hai.

Exam mein kya likhna hai?

Agar question aaye **"Explain Arithmetic Circuit"**, to ye structure follow karo:

Definition → Block Diagram → Components → Operation Table → Examples → Applications

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2. Shift Microoperation

Definition

Shift Microoperation ek microoperation hai jisme register ke binary bits ko **left ya right shift** kiya jata hai.

Example:

Before: 10110010

After: 01100100

Yahan bits shift hui hain.

Shift microoperations mainly **data manipulation** aur arithmetic operations mein useful hoti hain.

Types of Shift Microoperations

Mainly 3 types hain:

1. **Logical Shift**
2. **Circular Shift**
3. **Arithmetic Shift**

2.1 Logical Shift

Logical shift mein bits ko left ya right move kiya jata hai aur jo empty position banti hai usme generally **0** insert hota hai.

Logical Shift Left

Bits left side move karti hain.

Example:

Original: 10110110

Left Shift:

01101100

Leftmost bit bahar nikal jaati hai aur rightmost position par **0** aata hai.

Symbol

SHL / LSL

Logical Shift Right

Bits right side move karti hain.

Example:

Original: 10110110

Right Shift:

01011011

Rightmost bit bahar nikal jaati hai aur leftmost position par **0** aata hai.

Symbol

SHR / LSR

2.2 Circular Shift

Circular shift ko **Rotate operation** bhi kaha jata hai.

Isme jo bit ek side se bahar nikalti hai, woh doosri side se wapas enter kar jaati hai.

Circular Shift Left

Example:

Original: 10110010

Rotate Left:

01100101

Leftmost bit 1 rightmost position par aa gayi.

Circular Shift Right

Example:

Original: 10110010

Rotate Right:

01011001

Rightmost bit 0 leftmost position par aa gayi.

Important Point

Logical Shift → Empty position mein 0

Circular Shift → Removed bit doosri side se wapas

2.3 Arithmetic Shift

Arithmetic shift mainly **signed binary numbers** ke liye use hota hai.

Isme sign bit ko preserve karna important hota hai.

Usually **MSB (Most Significant Bit)** sign bit hoti hai.

Arithmetic Shift Right

Right shift ke time sign bit ko preserve kiya jata hai.

Example:

Original: 11010010

Arithmetic

Right Shift: 11101001

Leftmost sign bit 1 preserve hui.

Arithmetic Shift Left

Arithmetic left shift mein bits left move hoti hain aur right side mein 0 enter hota hai.

Example:

Original: 00110100

Arithmetic

Left Shift: 01101000

Shift Operations — Quick Comparison

Type	Main Feature
Logical Shift	Empty position mein 0
Circular Shift	Bahar nikli bit opposite side se enter
Arithmetic Shift	Sign bit ko preserve karta hai

Uses of Shift Microoperations

- Multiplication/division by powers of 2
- Data manipulation
- Bit-level operations
- Serial data transfer
- ALU operations

Easy Trick

Logical → 0

Circular → Rotate

Arithmetic → Sign preserve

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3. Hardwired Control Unit

Definition

Hardwired Control Unit ek control unit hai jisme control signals generate karne ke liye **fixed hardware circuits**, logic gates, flip-flops, decoders etc. use hote hain.

Isme control logic hardware mein permanently designed hoti hai.

Simple Definition for Exam

Hardwired Control Unit is a control unit in which control signals are generated by using fixed hardware logic circuits.

Working of Hardwired Control Unit

CPU ko instruction execute karne ke liye different control signals ki need hoti hai.

Hardwired Control Unit:

1. Instruction ko receive karta hai.
2. Opcode ko decode karta hai.
3. Timing signals ko consider karta hai.
4. Required control signals generate karta hai.
5. CPU ke different components ko control karta hai.

Main Inputs

Hardwired Control Unit ko mainly:

- **Opcode**
- **Timing signals**
- **Status/condition flags**

milte hain.

In inputs ke basis par control signals generate hote hain.

Hardwired Control Unit Diagram

Exam mein aap simple diagram is tarah draw kar sakte ho:

Instruction Register

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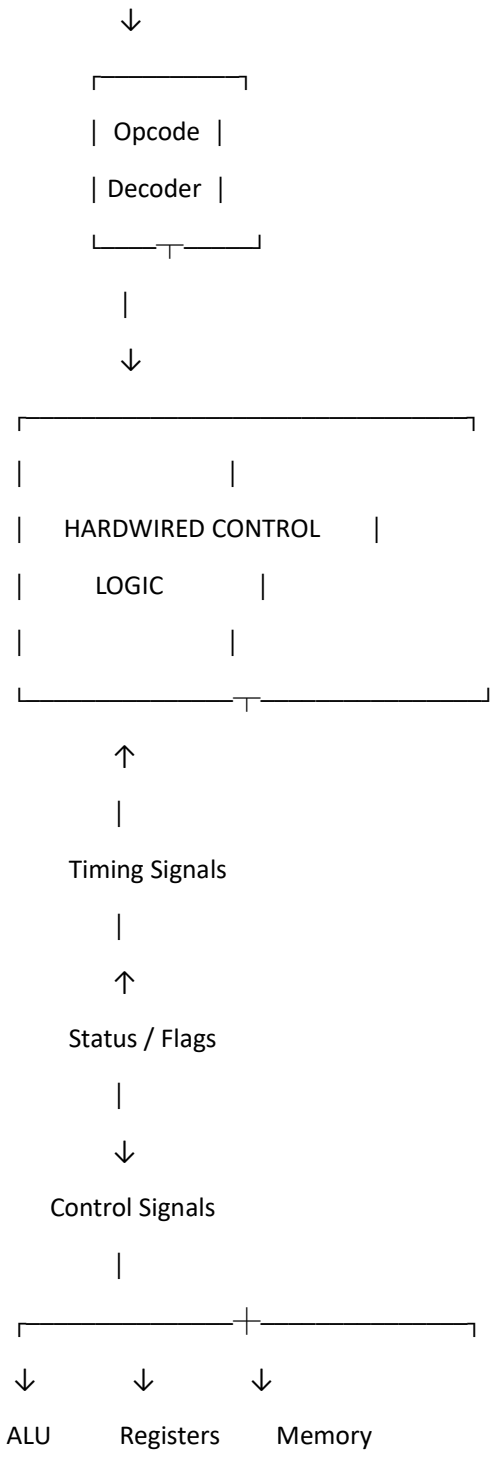


Diagram Explanation

Opcode Decoder:
Instruction ke opcode ko decode karta hai.

Timing Signals:
Batate hain ki operation kis timing par perform karna hai.

Status/Flags:
CPU ki current condition provide karte hain.

Control Logic:
In sab inputs ko process karke required control signals generate karta hai.

Advantages of Hardwired Control Unit

1. **Fast operation**
   Control signals directly hardware se generate hote hain.
2. **High speed**
   Extra microprogram memory access ki need nahi hoti.
3. **Efficient for simple instruction sets**
   Simple instructions ke liye suitable hai.
4. **No control memory required**
   Microinstructions store karne ke liye separate control memory nahi chahiye.

Disadvantages of Hardwired Control Unit

1. **Design complex ho sakta hai.**
2. **Modify karna difficult hota hai.**
3. New instructions add karna difficult ho sakta hai.
4. Large/complex instruction sets ke liye suitable nahi hota.
5. Hardware design change karna expensive ho sakta hai.

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4. Microprogrammed Control Unit

Definition

Microprogrammed Control Unit mein control signals ko generate karne ke liye **microinstructions** ka use hota hai.

Ye microinstructions ek special memory called **Control Memory** mein stored hoti hain.

Simple Definition

Microprogrammed Control Unit is a control unit in which control signals are generated by executing microinstructions stored in control memory.

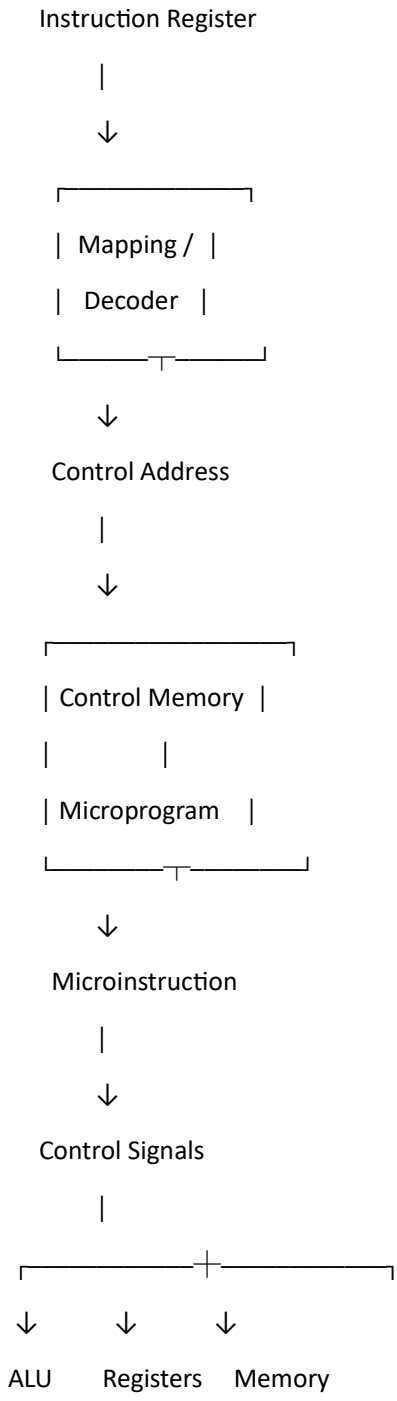
Microprogram

Microprogram instructions ka ek sequence hota hai jo kisi machine instruction ko execute karne ke liye required control signals generate karta hai.

Har small instruction ko **microinstruction** kaha ja sakta hai.

Multiple microinstructions milkar ek **microprogram** banati hain.

Basic Structure



Working of Microprogrammed Control Unit

Step 1

Instruction **Instruction Register** mein stored hoti hai.

Step 2

Instruction ka opcode identify/decode hota hai.

Step 3

Opcode ke basis par corresponding microprogram ka starting address find kiya jata hai.

Step 4

Control memory se microinstruction read hoti hai.

Step 5

Microinstruction required **control signals** generate karti hai.

Step 6

Next microinstruction ka address determine hota hai.

Ye process instruction complete hone tak continue hota hai.

Control Memory

Control Memory ek special memory hoti hai jisme microinstructions/microprograms stored hote hain.

Ye generally main memory se different hoti hai.

Main components

- Control Memory
- Control Address Register
- Microinstruction Register
- Sequencing logic

Advantages of Microprogrammed Control Unit

1. **Flexible** hota hai.
2. Modify karna easy hota hai.
3. New instructions add karna comparatively easy hota hai.
4. Complex instruction sets ke liye suitable hai.
5. Control logic ko microprogram change karke modify kiya ja sakta hai.
6. Design relatively systematic hota hai.

Disadvantages

1. Hardwired control unit se generally **slower** hota hai.
2. Control memory required hoti hai.
3. Microinstructions ko fetch karne mein additional time lagta hai.
4. Extra memory aur hardware required hota hai.

Hardwired vs Microprogrammed Control Unit

Hardwired	**Microprogrammed**
Fixed hardware logic use karta hai	Microinstructions use karta hai
Faster	Comparatively slower
Modify karna difficult	Modify karna easier
Control memory required nahi	Control memory required
Simple instruction sets ke liye suitable	Complex instruction sets ke liye suitable
Hardware based	Microprogram based

Easy Trick

Hardwired = Fast but Fixed

Microprogrammed = Flexible but Comparatively Slow

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5. Address Sequencing

Definition

Address Sequencing ka matlab hai control memory mein next **microinstruction ka address determine karna**.

Microprogrammed Control Unit mein ek microinstruction execute hone ke baad system ko decide karna hota hai ki **next microinstruction kahan se read karni hai**.

Isi process ko **Address Sequencing** kehte hain.

Why Address Sequencing is Needed?

Ek machine instruction ko execute karne ke liye ek se zyada microinstructions required ho sakti hain.

Example:

Machine Instruction

↓

Microinstruction 1

↓

Microinstruction 2

↓

Microinstruction 3

↓

Microinstruction 4

Har step ke baad next microinstruction ka address determine karna zaroori hai.

Sources of Next Address

Next microinstruction ka address different sources se aa sakta hai.

1. Incremented Address

Current microinstruction ke address ko increment karke next address obtain kiya ja sakta hai.

Example:

Current Address = 100

Next Address = 101

Ye sequential execution ke liye use hota hai.

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2. Branch Address

Kabhi-kabhi microprogram ko kisi different location par jump karna hota hai.

Is case mein branch address use hota hai.

Example:

Microinstruction 100

↓

↓ Branch

↓

Microinstruction 150

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3. Opcode Mapping

Machine instruction ke **opcode** ke basis par corresponding microprogram ka starting address determine kiya ja sakta hai.

Example:

ADD instruction → Microprogram starts at 200

LOAD instruction → Microprogram starts at 300

STORE instruction → Microprogram starts at 400

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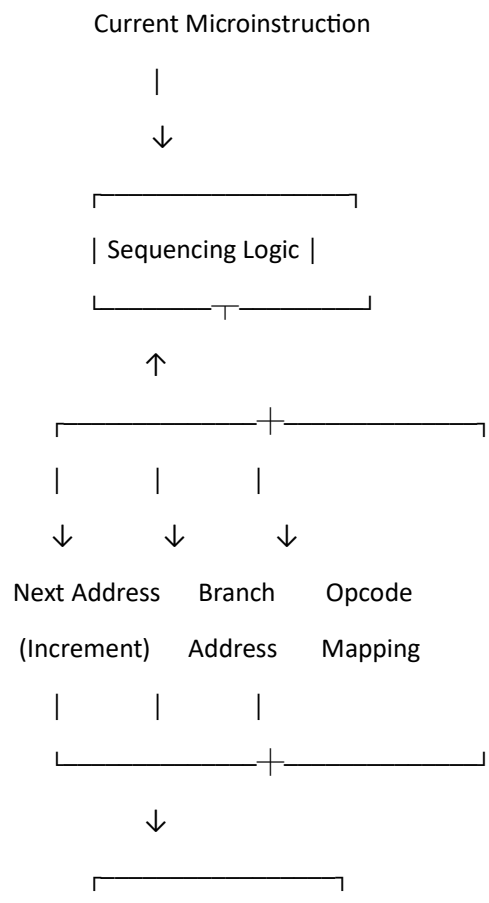
4. Subroutine Address

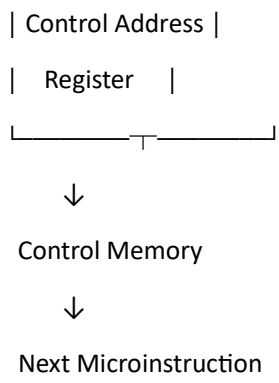
Agar same sequence of microinstructions multiple instructions mein required ho, to us sequence ko ek **microprogram subroutine** ke form mein store kiya ja sakta hai.

Required hone par control us subroutine ke address par jump karta hai.

Address Sequencing Diagram

Exam mein simple diagram:





Address Sequencing — Main Methods

Method	Meaning
Sequential	Current address ko increment karke next address
Branch	Different address par jump
Opcode Mapping	Opcode ke basis par starting address
Subroutine	Common microprogram sequence ko call karna